

Chapter 2: Vector Models and Text Preprocessing (Part 2)

Dr. BENDIABDALLAH M.H

University Belhadj BOUCHAIB – Ain Temouchent

February 14, 2025

Outline

- Tokenization
- Stopwords
- Stemming & Lemmatization

What is Tokenization?

- Splitting text into smaller units (tokens)
- Tokens can be words, subwords, or characters
- Fundamental for tasks like text classification and translation

Basic String Splitting

- Most languages offer a simple `split()` function.
- Example:

`"I like cats".split()` $\rightarrow$ `["I", "like", "cats"]`

- Simple but effective for many tasks.

Why Tokenization Matters

- Impacts vocabulary size and model complexity.
- Affects downstream tasks by preserving or losing context.
- **Key:** Good tokenization improves model performance.

Stopwords: What and Why?

- **Stopwords** are common words (e.g., "the", "and") that add little meaning.
- Removing them:
 - Reduces vocabulary size.
 - Lowers computational cost.
 - Helps highlight important words.

Why Reduce Word Variants?

- Variants like run, ran, running are treated as different tokens.
- This increases dimensionality and dilutes semantic similarity.
- Goal: Collapse words to their base form (root or lemma).

Lemmatization: A Refined Method

- Uses dictionaries and grammatical rules.
- Requires knowing the part-of-speech (POS) of the word.
- Example with WordNet Lemmatizer (NLTK):

Parts of Speech (POS)

- POS tags classify words.
- Accurate POS tagging improves lemmatization results.

Real-World Applications

- **Search Engines:** Normalize query terms
- **Online Ads:** Match keywords across variations.
- **Text Classification:** Reduce noise by collapsing word forms.